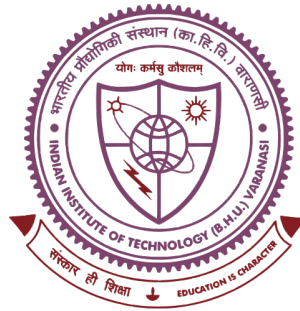


Spatiotemporal Forecasting of Solar Magnetic Winding-Flux Maps with Convolutional LSTMs

Toward Structure-Aware Solar Flare Prediction



Thesis submitted in partial fulfilment for the Award of
INTEGRATED DUAL DEGREE (B.TECH. + M.TECH.)
in **ENGINEERING PHYSICS**

by

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VARANASI – 221 005

May 2026

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The author wishes to express sincere gratitude to Dr. Abhishekh Kumar Srivastava, Department of Physics, IIT (BHU) Varanasi, for invaluable guidance, constant encouragement and the freedom to pursue this line of research throughout the course of this work. His insights on the physics of magnetic winding flux and on the methodology of spatiotemporal forecasting shaped the direction of the entire investigation.

Thanks are due to the faculty of the Department of Physics, IIT (BHU) Varanasi, for the rigorous coursework that built the foundation on which this thesis stands. The Department's commitment to integrating physics with contemporary computational methods made it possible to take up a topic that sits at the boundary between heliophysics, signal processing, and modern machine learning.

The data used in this work are derived from vector magnetograms produced by the Helioseismic and Magnetic Imager (HMI) aboard the Solar Dynamics Observatory (SDO); the magnetic winding-flux cubes were computed from these magnetograms within the present work, following the representation detailed in Chapter 2.

Finally, the author thanks family and friends for their patience and support during the long training runs and many late nights of experiment monitoring.

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Abstract

Solar flares are sudden releases of magnetic energy in the Sun’s atmosphere whose radiation and ejecta drive space-weather hazards at Earth. Forecasting them from the observable state of a source active region remains an open problem, made hard by the rarity of large events and by the fact that the governing coronal currents are seen only through their photospheric footprint. Magnetic *winding flux*—a signed, topological measure of how field lines wrap around one another, concentrated along polarity inversion lines—has been proposed as a physically motivated precursor, but existing work reduces the winding map to a single spatial integral before forecasting, cancelling the signed, localised structure that carries the signal.

This thesis takes a first step toward using the winding-flux map without that reduction. A Convolutional LSTM (ConvLSTM) encoder–forecaster is trained to predict the short-term evolution of the two-dimensional *accumulated total* winding field of an active region—the physically meaningful measure of its accumulated magnetic topology—from a ten-frame history, under a strict leave-one-active-region-out protocol. The total winding builds up gradually and is strongly persistent, and the model reproduces its large-scale evolution on held-out active regions across a 48-minute horizon. Skill is reported honestly against a persistence baseline—the model improves on persistence by about ten per cent—and the role of the spatial low-pass is stated plainly rather than hidden. The work establishes both a working spatiotemporal pipeline for winding-flux maps and a clear-eyed account of what such forecasting can and cannot deliver, motivating the structure-aware direction for subsequent flare-prediction research.

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Symbols and Abbreviations

Abbreviations

AR	Active Region (a region of strong, structured magnetic field on the solar surface)
SDO	Solar Dynamics Observatory
HMI	Helioseismic and Magnetic Imager (onboard SDO)
SHARP	Spaceweather HMI Active Region Patch
HARP	HMI Active Region Patch
GOES	Geostationary Operational Environmental Satellite
PIL	Polarity Inversion Line
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
ConvLSTM	Convolutional LSTM
MAE	Mean Absolute Error
TSS	True Skill Statistic
CSI	Critical Success Index
asinh	Inverse hyperbolic sine (sign-preserving amplitude transform)

Symbols

$w(\mathbf{r}, t)$	Per-frame winding-flux density at pixel $\mathbf{r}$, time t
$W(\mathbf{r}, t)$	Accumulated total winding, $W[t] = \sum_{\tau \leq t} w[\tau]$ (forecasting target)
s	Softening constant of the asinh transform, $s = 10^3$
t_{in}	Number of input (history) frames per window
t_{out}	Number of forecast (future) frames per window
T	Total frames per window, $T = t_{\text{in}} + t_{\text{out}}$
i, f, g, o	ConvLSTM input, forget, candidate and output gates
h_t, c_t	ConvLSTM hidden and cell state maps at time t
$\hat{x}_{t+1}$	Model prediction of the frame at time $t+1$
Mm	Megametre, 10^6 m (used for physical pixel scale)

Preface

This thesis investigates spatiotemporal deep-learning models for the magnetic winding-flux field of solar active regions, derived from Solar Dynamics Observatory (SDO) Helioseismic and Magnetic Imager (HMI) observations. The work was carried out in the Department of Physics, Indian Institute of Technology (Banaras Hindu University), Varanasi, as part of the Integrated Dual Degree (B.Tech. + M.Tech.) programme in Engineering Physics.

Subject. The subject is the design, training and honest evaluation of a Convolutional LSTM (ConvLSTM) encoder–forecaster that predicts the short-term evolution of the two-dimensional winding-flux map of an active region, as a step toward structure-aware solar flare forecasting.

Scope. The thesis is organised as three chapters. Chapter 1 introduces solar flares, the space-weather forecasting problem, and the current data-driven landscape, and identifies the gap addressed here: existing work integrates the winding-flux map to a single scalar, discarding its spatial structure. Chapter 2 specifies the data representation, the ConvLSTM architecture, and the leave-one-active-region-out training procedure. Chapter 3 reports the forecasting results and analyses them against a persistence baseline, distinguishing genuine predictive skill from the persistence afforded by a smoothly evolving field.

A note on honesty. Because the accumulated total winding is strongly persistent, every forecast is reported against a persistence baseline, and the effect of the spatial low-pass is stated plainly in the text rather than obscured. No result in this thesis is presented as stronger than the evidence supports.

Manik Sharma

Varanasi

Chapter 1

Solar Flares and the Forecasting Problem

1.1 Solar flares and space weather

A solar flare is a sudden, localised release of magnetic energy in the Sun’s atmosphere, radiating across the electromagnetic spectrum on timescales of minutes. Flares originate in *active regions* (ARs): concentrations of strong, structured magnetic field that pierce the photosphere as sunspot groups. The energy that powers a flare is not thermal but magnetic—it is stored in electric currents that thread the coronal field and is liberated when the field reconnects into a lower-energy configuration. The largest events are frequently accompanied by coronal mass ejections, expelling magnetised plasma into interplanetary space.

The practical importance of flares lies in their terrestrial impact, collectively termed *space weather*. Flare radiation and the energetic particles and plasma that follow can disturb the ionosphere and disrupt high-frequency radio communication and satellite navigation, induce currents in power grids, raise radiation doses for astronauts and polar-route aircraft, and shorten the operational lifetime of spacecraft. Because the radiative onset of a flare arrives at Earth in a matter of minutes, useful mitigation depends on *forecasting*—estimating the probability of a significant event *before* it occurs, from the observable state of the source active region.

Flares are classified by their peak soft X-ray flux in the 1–8 Å band measured by the GOES satellites, in a logarithmic scheme with classes A, B, C, M and X. Operationally the threshold of concern is the M class and above: $\geq M$ events are energetic enough to drive the impacts listed above, yet rare enough that reliable anticipation remains an open problem. Throughout this thesis, “a flare” without further qualification refers to a $\geq M$ event, and forecasting is framed on the active-region scale using photospheric magnetic data from the Helioseismic and Magnetic Imager (HMI) aboard the Solar Dynamics Observatory (SDO). Figure 1.1 shows the classification scale and the $\geq M$ target regime.

1.2 Why forecasting is hard

The difficulty of flare forecasting is fundamental rather than merely technical. Three properties conspire against it. First, the process is driven by the *coronal* magnetic field

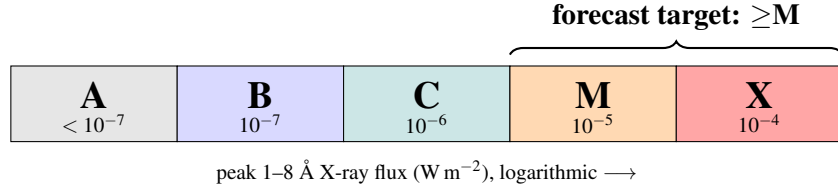


Figure 1.1: The GOES soft X-ray flare classification. Flares are graded A–X by their peak 1–8 Å X-ray flux, each class a factor of ten above the last. Operational concern—and the target of this work—is the $\geq M$ regime: energetic enough to drive space-weather impacts, yet rare enough that reliable anticipation remains difficult.

and its currents, whereas routine observations resolve only the *photospheric* field at the base of the atmosphere; the quantity that matters must be inferred from a projection of it. Second, flaring is governed by a threshold instability: an active region can store free energy for days and release it in seconds, so the mapping from a slowly evolving observable to a near-instantaneous event is weakly constrained. Third, $\geq M$ flares are *rare*. In any large catalogue the flaring minority is outnumbered by quiet examples by more than an order of magnitude, so a forecaster is judged not on raw accuracy—trivially maximised by always predicting “quiet”—but on class-balanced skill scores such as the True Skill Statistic (TSS), which reward correct anticipation of the rare positive class.

1.3 The current forecasting landscape

Modern data-driven flare forecasting was catalysed by the availability of uniform, machine-readable magnetic-field summaries of active regions. The Space-weather HMI Active Region Patch (SHARP) product distils each AR’s vector magnetogram into a handful of scalar parameters—total unsigned flux, total current helicity, twist, and related quantities. Bobra and Couvidat [1] showed that a support-vector machine trained on these SHARP scalars produces a competitive $\geq M$ forecaster, and established the SHARP feature vector as the de-facto baseline against which later work is measured.

Two broad strands of deep learning followed. The first replaces hand-designed scalars with features learned directly from the magnetogram *image*: Huang et al. [3] trained convolutional neural networks on line-of-sight magnetograms, letting the network discover spatial precursors rather than relying on pre-integrated summaries. The second exploits the *temporal* evolution of an active region: Liu et al. [4] fed sequences of SHARP parameter vectors to a Long Short-Term Memory (LSTM) network, capturing the build-up of free energy over the hours preceding an event, while operational systems such as Deep Flare Net [2] combine engineered features with deep architectures for daily probabilistic forecasts. The recurring lesson of this literature is that both the *spatial structure* of the field and its *temporal history* carry predictive information; the best systems try to use both.

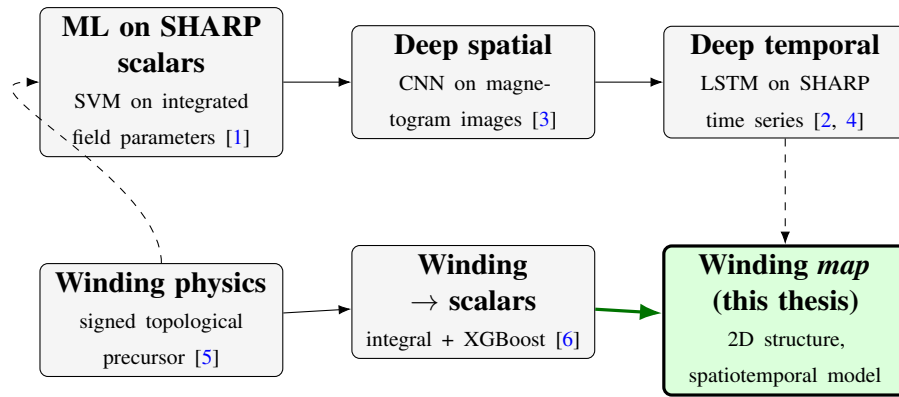


Figure 1.2: Where this thesis sits. Data-driven flare forecasting progressed from machine learning on integrated SHARP scalars to deep spatial and deep temporal models. In parallel, magnetic winding emerged as a physically motivated precursor, but existing work reduces the winding *map* to a single spatial integral, in which oppositely signed regions cancel. This thesis instead feeds the two-dimensional winding map, with its structure intact, to a spatiotemporal model.

Running alongside this forecasting work is a strand of solar physics that seeks a more physical descriptor of an active region’s readiness to flare. MacTaggart et al. [5] demonstrated that emerging active regions carry a systematic magnetic *winding*—a signed, topological measure of how field lines wrap around one another—providing direct evidence that twisted flux tubes create ARs. Magnetic winding flux is attractive precisely because it is rooted in the field’s topology rather than its raw strength, and it is the central physical quantity of this thesis. Figure 1.2 places these strands, and the present work, in a single picture.

1.4 Winding flux and the gap addressed by this thesis

Magnetic winding flux is a *signed* field defined pixel-by-pixel across an active region: it is positive where field lines wind one way and negative where they wind the other, and it is concentrated along polarity inversion lines, where flares originate. Existing work that uses winding reduces the map to a single number—its spatial integral—before forecasting. The most direct example is recent: Williams et al. [6] derive scalar winding features from active-region maps and feed them to gradient-boosted trees (XGBoost), obtaining competitive flare-prediction skill and thereby confirming that winding carries genuine predictive signal—yet the map is collapsed to a handful of numbers before the classifier ever sees it. This integration is convenient but lossy: because winding is signed, oppositely wound regions cancel in the spatial mean, discarding exactly the localised, sign-resolved structure that distinguishes a flare-productive configuration from a quiet one. Figure 1.3 makes the loss concrete: on a real active region the magnitude of the spatial mean is roughly twenty times smaller than the spatial standard deviation, so

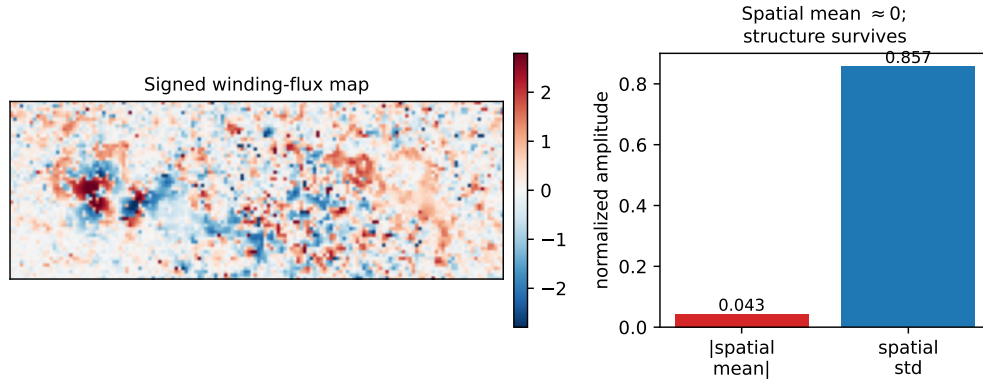


Figure 1.3: Why the spatial integral throws the signal away. Left: a signed winding-flux map of a real active region, with oppositely wound (red/blue) regions. Right: the magnitude of the spatial mean is only a small fraction of the spatial standard deviation—the positive and negative winding very nearly cancel, so a single integrated scalar discards the localised structure that the two-dimensional map retains.

integrating to a scalar discards almost all of the structure. To date, no study has fed the *two-dimensional* winding-flux map, with its spatial structure intact, to a deep model.

This thesis takes a first step into that gap. Rather than forecasting a flare label directly, it asks a more basic question about the winding-flux field itself: *given a short history of winding-flux maps of an active region, can a spatiotemporal deep model predict how the field evolves?* A model that can anticipate the field’s evolution has, in effect, learned the structure and dynamics that a downstream forecaster would need. The vehicle for this study is the Convolutional LSTM (ConvLSTM), a neural architecture that couples the spatial inductive bias of convolution with the temporal memory of an LSTM, and is therefore matched to the spatiotemporal nature of the data.

The remainder of the thesis is organised as follows. Chapter 2 develops the technical architecture: the winding-flux data and its representation, the ConvLSTM cell and the encoder–forecaster built from it, and the training procedure. Chapter 3 reports the forecasting results, states plainly what the model does and does not achieve, and analyses the outcome in light of the physical nature of the winding field—separating genuine predictive skill from the persistence that a smoothly evolving field affords for free.

Chapter 2

Technical Architecture

This chapter specifies the complete forecasting system: the winding-flux data and how it is represented (§2.1), the Convolutional LSTM cell and the encoder–forecaster assembled from it (§2.2–§2.3), and the training procedure (§2.4). Every design choice that affects the reported results—in particular the spatial low-pass and the choice of the *accumulated* winding field—is stated explicitly so that the outcome in Chapter 3 can be read without hidden assumptions.

2.1 Data and representation

Winding-flux datacubes. The input data are per-active-region datacubes derived from SDO/HMI observations. Each cube stores a single channel $\text{wind}[H, W, T]$ —the winding-flux density on an $H \times W$ spatial grid at T successive times—together with a vector of acquisition timestamps at a nominal 12-minute cadence. A small number of frames are missing; these are flagged by a sentinel timestamp of zero and are removed before any windowing, so that the model only ever sees physically contiguous sequences. The corpus used here comprises the available HARP cubes; one cube exhibiting pathological, noise-dominated values is excluded from every experiment.

The active-region corpus. The dataset comprises 27 active regions and 13,472 valid frames after sentinel removal, a median of 421 frames per region (Figure 2.2); the regions vary widely in size—from a few hundred to over a thousand pixels on a side—and in the morphology of their winding field (Figure 2.1). Each map is signed and bipolar, with opposed regions of positive and negative winding meeting along the internal boundaries of the active region. This diversity is the reason the evaluation is leave-one-active-region-out (§2.4): the model must generalise across regions of different scale and structure, not merely interpolate within one. Table 2.1 lists every region with its frame count and grid size. The corpus also carries the characteristic imprint of how the data are measured: regions observed near the solar limb suffer the *limb effect*, where line-of-sight foreshortening degrades the vector-field inversion and leaves part of the patch with unreliable, low-signal winding (visible in HARP 892 in Figure 2.1). Such near-limb passages are a known limitation of photospheric magnetography and bound the usable field of view for any method built on these maps.

Magnetic winding-flux maps across active regions (asinh-scaled)

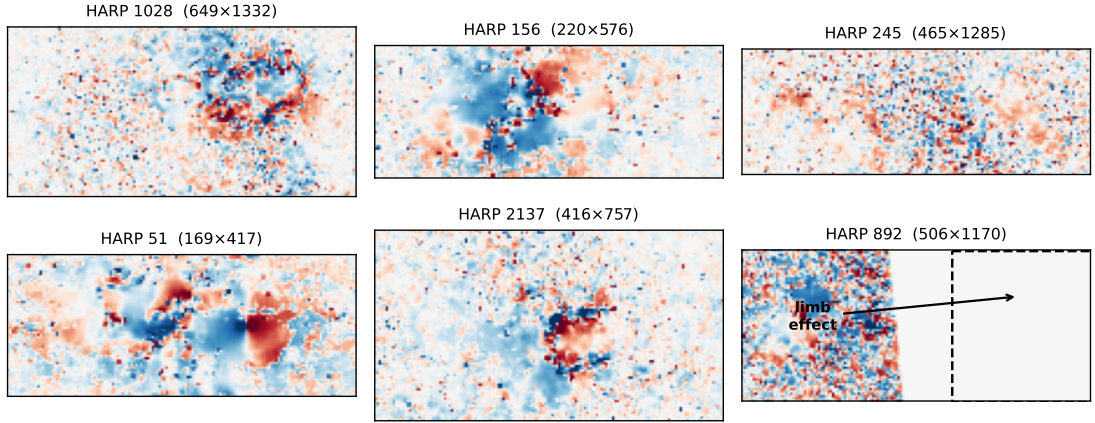


Figure 2.1: Magnetic winding-flux maps across the active-region corpus. One representative frame from each of six HARP regions, asinh-scaled, with grid dimensions given. The regions differ markedly in size and in the morphology of their signed (red/blue) winding structure, which is why the model is evaluated leave-one-active-region-out. HARP 892 (bottom right) illustrates the *limb effect*: when an active region lies close to the solar limb, the strong foreshortening of the line-of-sight geometry degrades the vector-field inversion, leaving a low-signal region (dashed box) where the winding is unreliable.

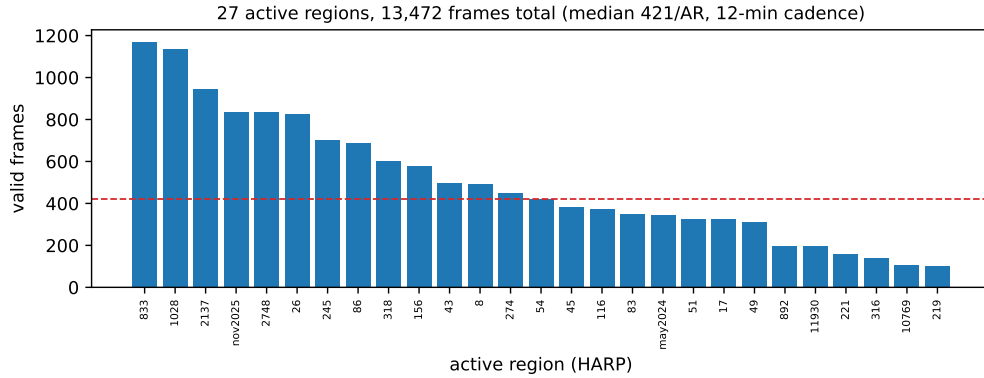


Figure 2.2: Valid frames per active region. The corpus holds 27 active regions and 13,472 frames at 12-minute cadence after sentinel removal (dashed line: median 421 frames per region). Temporal coverage varies by an order of magnitude across regions.

Table 2.1: The active-region corpus. Valid frames (after sentinel removal) and native grid size for each of the 27 HARP regions, at 12-minute cadence. HARP 11930 is the held-out evaluation region; HARP 1149 is excluded as noise-dominated. Totals: 13,472 frames, median 421 per region.

HARP	Frames	Grid	Role	HARP	Frames	Grid	Role
833	1169	507×976	train	45	382	125×323	train
1028	1136	649×1332	train	116	375	213×517	train
2137	946	416×757	train	83	347	126×283	train
nov2025	837	437×1042	train	may2024	342	440×884	train
2748	834	356×787	train	17	324	91×173	train
26	826	215×473	train	51	324	169×417	train
245	703	465×1285	train	49	309	239×495	train
86	687	566×981	train	11930	194	627×877	holdout
318	601	161×442	train	892	194	506×1170	train

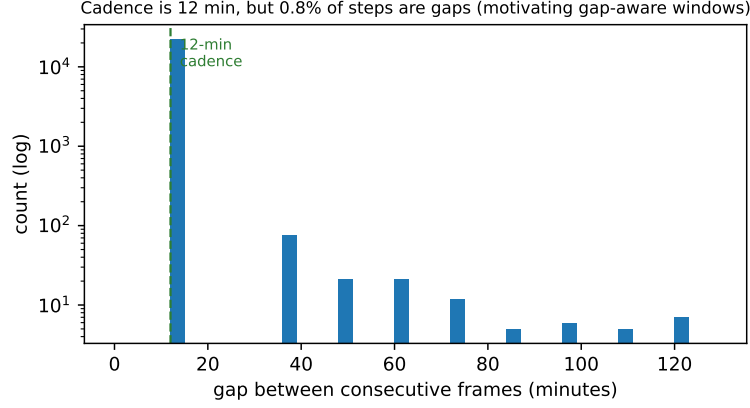


Figure 2.3: Frame-to-frame time gaps. The cadence is 12 minutes (green line) for the overwhelming majority of steps; a small tail (under one per cent) occurs at integer multiples of the cadence, where acquisitions are missing. These gaps motivate gap-aware windowing.

Cadence and gaps. The nominal cadence is 12 minutes, and the overwhelming majority of steps respect it exactly (Figure 2.3). A small fraction—under one per cent—are longer, appearing at integer multiples of the cadence where one or more acquisitions are missing. These gaps are why windowing must be *gap-aware*: a window whose ten input and four target frames are not all consecutive in real time would splice across a temporal discontinuity and present the model with a jump that never occurs in deployment. Every training and evaluation window used in this thesis is gap-free by construction.

The signed field and its dynamic range. Winding-flux density is a *signed* pseudoscalar with a very large dynamic range. Its distribution is sharply peaked at zero but heavy-tailed: the bulk of pixels lie near the origin while a few reach magnitudes of order 10^7 , several decades above the typical value (Figure 2.4, left). Training a network directly on such values is unstable—the rare extremes dominate the loss and the gradients. The field is therefore passed through a sign-preserving *inverse hyperbolic sine* (asinh) transform,

$$\tilde{w} = \operatorname{asinh}\left(\frac{w}{s}\right), \quad s = 10^3, \quad (2.1)$$

which is linear for $|w| \ll s$ and logarithmic for $|w| \gg s$ (Figure 2.4, centre). This compresses the extreme tail while preserving sign and small-signal structure, turning the raw heavy-tailed distribution into the well-behaved, roughly symmetric one of Figure 2.4 (right); the softening constant $s = 10^3$ was fixed by prior probing of the field’s amplitude distribution. The transformed field is then rescaled by a robust (99.5th percentile) amplitude and clipped to $[-1, 1]$ per cube, giving a normalised, zero-centred representation comparable across active regions.

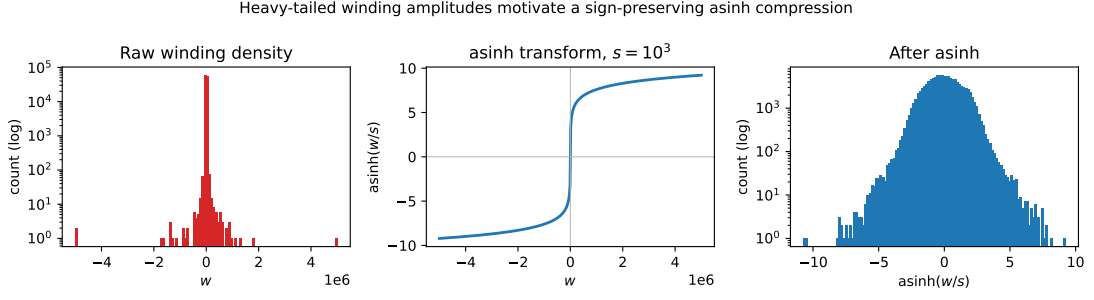


Figure 2.4: Heavy-tailed amplitudes motivate the asinh transform. Left: the raw winding-density distribution (log count) is sharply peaked with tails reaching $\sim 10^7$. Centre: the sign-preserving asinh transform of Eq. (2.1), linear near zero and logarithmic in the tails. Right: after the transform the distribution is well-behaved and roughly symmetric, suitable for training.

The accumulated total winding. The quantity this thesis forecasts is the *accumulated total* winding: the net magnetic winding injected into an active region up to time t , obtained by integrating the per-frame winding over time,

$$W[t] = \sum_{\tau \leq t} \text{wind}[\tau]. \quad (2.2)$$

This is the physically meaningful measure of an active region’s accumulated non-potential topology, and it is the natural forecasting target: it is smooth and strongly persistent from frame to frame, with a lag-of-one spatial correlation of about 0.98 that decays only gradually over hours (Figure 2.5). Its gradual build-up is exactly what a spatiotemporal model can learn to extrapolate. Unless stated otherwise, the model is trained and evaluated on the accumulated total W of Eq. (2.2), consistent with the total winding used throughout this project.

Spatial low-pass and downsampling. Before entering the network each frame is area-downsampled to a fixed width and convolved with a separable Gaussian kernel. This is a deliberate, disclosed choice, and the data motivate it directly. The winding field carries appreciable power at every spatial scale, right down to the pixel: its radially-averaged power spectrum falls off only slowly and never flattens into a noise floor (Figure 2.6). That fine-scale content is not, however, useful for forecasting—the same small scales that the spectrum shows to be energetic are the ones whose frame-to-frame evolution is incoherent (Figure 2.5), so they cannot be predicted. The low-pass discards this energetic-but-unpredictable detail in favour of the large-scale envelope, which evolves coherently, letting the model concentrate its capacity on the part of the field that can actually be forecast. The cost—loss of sub-envelope structure—is real and is acknowledged wherever results are reported. Figure 2.7 summarises the full pipeline from the raw winding cube to the gap-aware training windows.

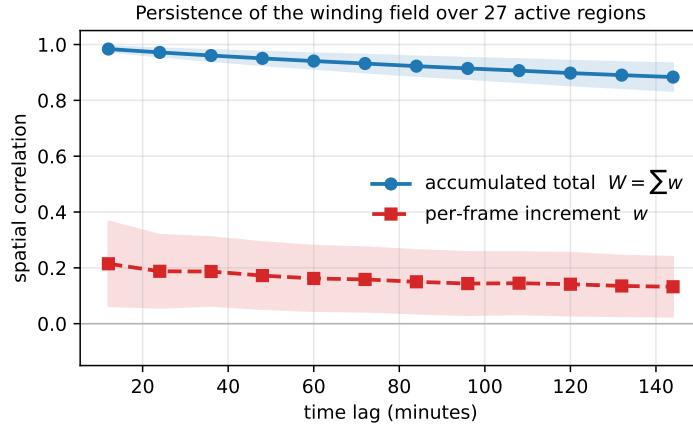


Figure 2.5: The accumulated total winding is smooth and strongly persistent. Spatial correlation between winding frames separated by a given time lag, averaged over the active-region corpus (shaded band: one standard deviation across regions). The accumulated total (blue) remains strongly correlated over hours, decaying only gradually—the property that makes it a well-posed forecasting target. The per-frame increment (red) is shown for contrast. Because the total is so persistent, its forecast skill must be read against a persistence baseline, as it is throughout Chapter 3.

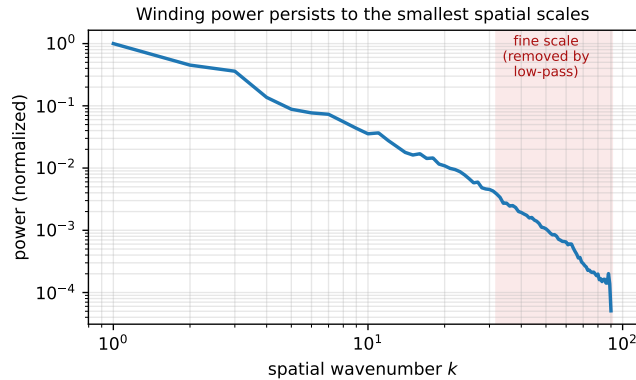


Figure 2.6: Winding power persists to the smallest spatial scales. Radially-averaged spatial power spectrum of the asinh winding field, averaged over active regions and normalised to the largest scale. Power falls off only gradually with wavenumber and never reaches a flat noise floor: there is real structure at every scale. Combined with the temporal incoherence of those small scales (Figure 2.5), this justifies removing the fine-scale band (shaded) by low-pass before forecasting.

2.2 The Convolutional LSTM cell

The temporal backbone of the model is the Convolutional LSTM (ConvLSTM) of Shi et al. [7], which adapts the Long Short-Term Memory unit of Hochreiter and Schmidhuber [8] to spatial data. A standard LSTM maintains a memory cell gated by learned nonlinearities but treats its input as a flat vector, destroying spatial structure. The ConvLSTM keeps the same gating logic but replaces every matrix multiplication with a two-dimensional convolution, so that the hidden state h and cell state c are *feature maps*

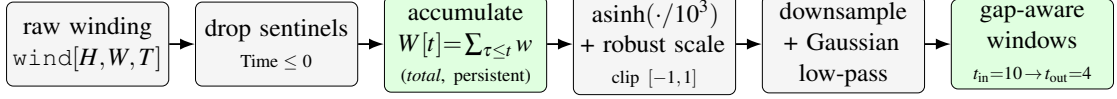


Figure 2.7: Winding-flux data pipeline. Missing frames (sentinel timestamps) are removed, and the winding is accumulated over time into the persistent *total* winding $W[t]$ (Eq. 2.2), the forecasting target. The total is passed through the sign-preserving asinh transform of Eq. (2.1) and a robust rescale, spatially downsampled and low-passed (§2.1), and finally cut into gap-aware history/future windows for training (§2.4).

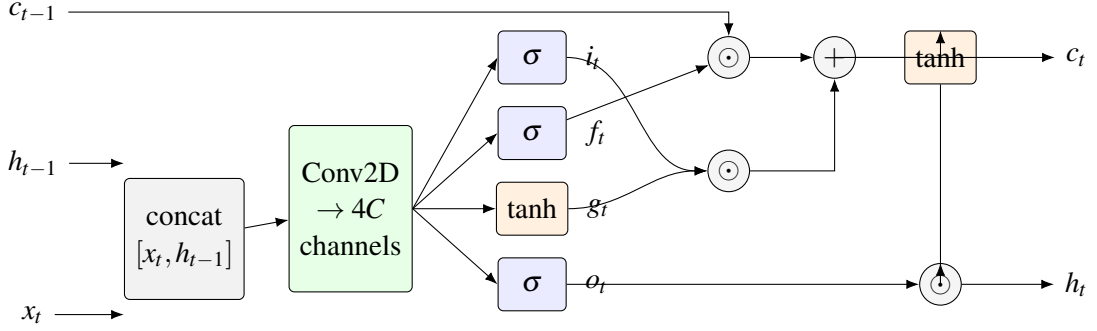


Figure 2.8: The Convolutional LSTM cell. A single 2D convolution over the channel-concatenated input and previous hidden state $[x_t, h_{t-1}]$ produces all four gate maps at once (input i_t , forget f_t , candidate g_t , output o_t ; Eq. 2.3). The cell state updates as $c_t = f_t \odot c_{t-1} + i_t \odot g_t$ and the hidden state as $h_t = o_t \odot \tanh(c_t)$ (Eq. 2.4). All states are feature maps, so spatial structure is preserved through the recurrence.

rather than vectors, and spatial locality is preserved throughout the recurrence.

At each timestep the current input frame x_t and the previous hidden state h_{t-1} (both feature maps) are concatenated along the channel axis and passed through a single convolution whose output is split into four gate maps—input i , forget f , candidate g and output o :

$$\begin{aligned} i_t &= \sigma(W_i * [x_t, h_{t-1}] + b_i), & f_t &= \sigma(W_f * [x_t, h_{t-1}] + b_f), \\ g_t &= \tanh(W_g * [x_t, h_{t-1}] + b_g), & o_t &= \sigma(W_o * [x_t, h_{t-1}] + b_o), \end{aligned} \quad (2.3)$$

where $*$ denotes convolution and σ the logistic sigmoid. The cell and hidden states update as

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t, \quad h_t = o_t \odot \tanh(c_t), \quad (2.4)$$

with $\odot$ the elementwise product. Computing all four gates with one convolution is an efficiency choice; using “same” padding keeps the spatial resolution fixed through the recurrence. Two implementation details matter for stable training: the forget-gate bias b_f is initialised to $+1$, so that the cell defaults to *remembering* its state early in training and gradients propagate cleanly through time; and normalisation between stacked layers uses group normalisation, which carries no running statistics and is therefore safe to reuse across the unrolled decoding loop. Figure 2.8 traces the internal dataflow of a single cell.

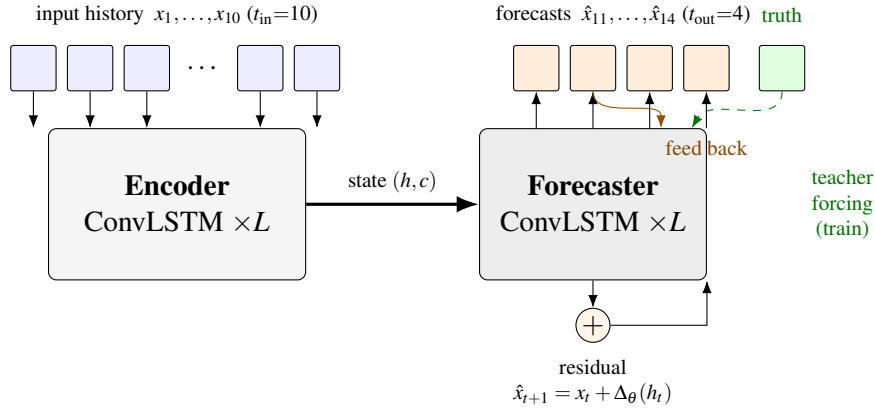


Figure 2.9: The encoder-forecaster model. A stack of L ConvLSTM layers encodes the $t_{in}=10$ history frames into the recurrent state (h, c) ; a second stack then rolls the state forward for $t_{out}=4$ steps. Each step predicts a *residual* $\hat{x}_{t+1} = x_t + \Delta_{\theta}(h_t)$ (Eq. 2.5), so a null network falls back to persistence, and feeds its own prediction back as the next input. During training the fed-back frame may be replaced by the ground truth with probability equal to the teacher-forcing ratio (dashed); at evaluation the model runs purely autoregressively.

2.3 Encoder-forecaster model

The full model is an *encoder-forecaster*: a stack of ConvLSTM layers first reads the observed history, and a second stack then generates the future frames autoregressively. Given an input clip of $T_{in} = 10$ frames, the encoder applies Eq. (2.3)–(2.4) layer by layer and frame by frame, accumulating the sequence into the hidden and cell states of every layer. No output is produced during encoding; its sole purpose is to summarise the history into the recurrent state.

Decoding then rolls the state forward for $T_{out} = 4$ steps. At each step the decoder stack advances the recurrent state and a 1×1 convolution maps the top hidden map to a single-channel prediction. Crucially the model predicts a *residual*: the output is the previous frame plus a learned increment,

$$\hat{x}_{t+1} = x_t + \Delta_{\theta}(h_t), \quad (2.5)$$

so that a do-nothing network falls back to *persistence*—copying the last frame—rather than collapsing to zero. This is the appropriate inductive bias for a strongly persistent field and, as Chapter 3 makes clear, it also sets the correct baseline against which any genuine skill must be measured. The predicted frame is fed back as the next decoder input, so errors compound along the roll-out exactly as they would at deployment. Figure 2.9 shows the encoder, the state hand-off, and the autoregressive decoder together.

During training the fed-back frame may instead be the ground-truth frame with a tunable probability—*teacher forcing*—which stabilises early optimisation by anchoring the roll-out to real data; at evaluation teacher forcing is switched off and the model runs

purely on its own predictions.

2.4 Training procedure

The model is optimised with Adam and a per-frame regression loss (a smoothed L_1 / Huber criterion) between the predicted and true future frames, summed over the T_{out} horizon. Gradients are clipped in norm to curb the occasional large update produced by rare high-amplitude frames, and the learning rate follows a cosine decay across the training budget.

Evaluation follows a strict *leave-one-active-region-out* protocol. Because successive 12-minute frames within a cube are highly autocorrelated, any split that mixes frames from the same active region between training and test leaks information and inflates the apparent score. The model is therefore trained on the pooled windows of all active regions except one, and evaluated only on the held-out region it has never seen. All figures and metrics in Chapter 3 are reported on such held-out regions, together with the persistence baseline of Eq. (2.5), so that model skill is always stated relative to the trivial forecast. Table 2.2 collects the complete data, model, and training configuration in one place.

Table 2.2: Data, model, and training configuration. The complete set of settings used for the leave-one-active-region-out experiment of Chapter 3.

Group	Setting	Value
Data	forecasting target	accumulated total winding W
	input / output horizon	$t_{\text{in}} = 10 / t_{\text{out}} = 4$ frames (120 / 48 min)
	asinh softening	$s = 10^3$; robust clip to $[-1, 1]$
	spatial low-pass	Gaussian $\sigma = 1.5$, grid 88×132
	windowing	gap-aware, stride 3
Model	architecture	ConvLSTM encoder–forecaster
	layers / hidden width	3 / 96
	kernel	3×3 , residual, group norm
	decoder	autoregressive, 1×1 conv head
	parameters	3.33 M
Training	optimiser	Adam, learning rate 10^{-3}
	schedule	cosine decay over 48 epochs
	batch size	4
	gradient clipping	max norm 1.0
	loss	smooth- L_1 (Huber)
	evaluation	leave-one-AR-out vs. persistence

Chapter 3

Results and Analysis

This chapter reports the forecasting experiment and reads its outcome honestly. The model of Chapter 2 is trained and evaluated under the leave-one-active-region-out protocol (§3.1); the forecasts on a held-out region are shown and quantified (§3.2); and the result is then interpreted, separating the genuine skill of the model from the persistence that a smoothly evolving field affords for free (§3.3).

3.1 Experimental setup

The encoder–forecaster is trained on the pooled gap-aware windows of 26 active regions and evaluated on a single held-out region, HARP 11930, that it never sees during training. The noisy region flagged in §2.1 is excluded from every split. Each cube contributes only windows whose ten input and four target frames are consecutive in real time at the nominal 12-minute cadence, so no window straddles an acquisition gap; this yields 2,784 training windows and 61 held-out windows. All frames are the accumulated total winding of Eq. (2.2), asinh-normalised and low-passed onto a fixed 88×132 grid.

The model is a three-layer ConvLSTM encoder–forecaster (3.3 million parameters, hidden width 96), trained for 48 epochs with Adam, a cosine learning-rate decay, and gradient-norm clipping, forecasting $t_{\text{out}} = 4$ frames (48 minutes) from $t_{\text{in}} = 10$ history frames (120 minutes). The reference against which it is judged is the *persistence* forecast—copying the last observed frame forward—which is the natural zero-skill baseline for a strongly autocorrelated field and the fallback the residual parameterisation of Eq. (2.5) is built around. Skill is defined as the fractional reduction in mean absolute error relative to persistence,

$$\text{skill} = 1 - \frac{\text{MAE}_{\text{model}}}{\text{MAE}_{\text{persistence}}}. \quad (3.1)$$

3.2 Forecasting the held-out active region

Headline result. On the held-out region HARP 11930—never seen in training—the ConvLSTM forecasts the accumulated winding field over a 48-minute horizon with a mean absolute error of **0.0106**, against **0.0117** for the persistence baseline: a skill of

+**9.3%** by Eq. (3.1), measured over all 61 gap-aware windows of the region. The model therefore generalises across active regions and improves on the trivial forecast. Table 3.1 states these numbers together.

Table 3.1: Forecast accuracy on the held-out active region. Mean absolute error (normalised units) over all 61 gap-aware windows of HARP 11930, for the ConvLSTM and for the persistence baseline, with the skill of Eq. (3.1). The model was trained only on the other 26 regions.

Forecaster (held-out HARP 11930, 48-min horizon)	MAE
Persistence (copy last observed frame)	0.0117
ConvLSTM encoder–forecaster (this work)	0.0106
Skill over persistence	+9.3%

Frame-by-frame verification. A forecast is judged not by how good the prediction looks but by how it differs from the truth. Figure 3.1 places the two side by side at every lead time: the top row is the prediction, the middle row the observation at the same instant, and the bottom row their difference. Prediction and truth are near-identical at the large scale. The residuals, on the expanded scale of the bottom row, are small (a few per cent of the field amplitude) and—tellingly—*structured*: they are not spread uniformly but concentrate in fine specks along the boundaries between the opposed winding lobes, precisely where the field has its sharpest gradients. This is the fine-scale content that §2.1 identified as temporally incoherent: the model reproduces the smooth envelope and errs only on the unpredictable detail, and the error grows frame by frame as that detail decorrelates.

How error grows with lead time. Figure 3.2 traces the mean absolute error, averaged over all 61 held-out windows, as a function of lead time, for the model and for persistence. Both grow as the field decorrelates, but their relationship is informative: at +12 minutes the two coincide—over a single step the accumulated field barely changes, so nothing beats copying it—while at every later step the model pulls below persistence, the margin widening to +48 minutes. The model has therefore learned genuine short-term dynamics, and its advantage is largest exactly where persistence degrades fastest.

Qualitative agreement across the sequence. Finally, Figure 3.3 surveys three windows drawn from different days across the held-out sequence, showing the input, the four forecasts, and the observed frame. Sampled evenly, these span the range of per-window error (mean absolute error from 0.028 down to 0.007); across all of them the large-scale organisation of the winding field is reproduced over the full horizon.

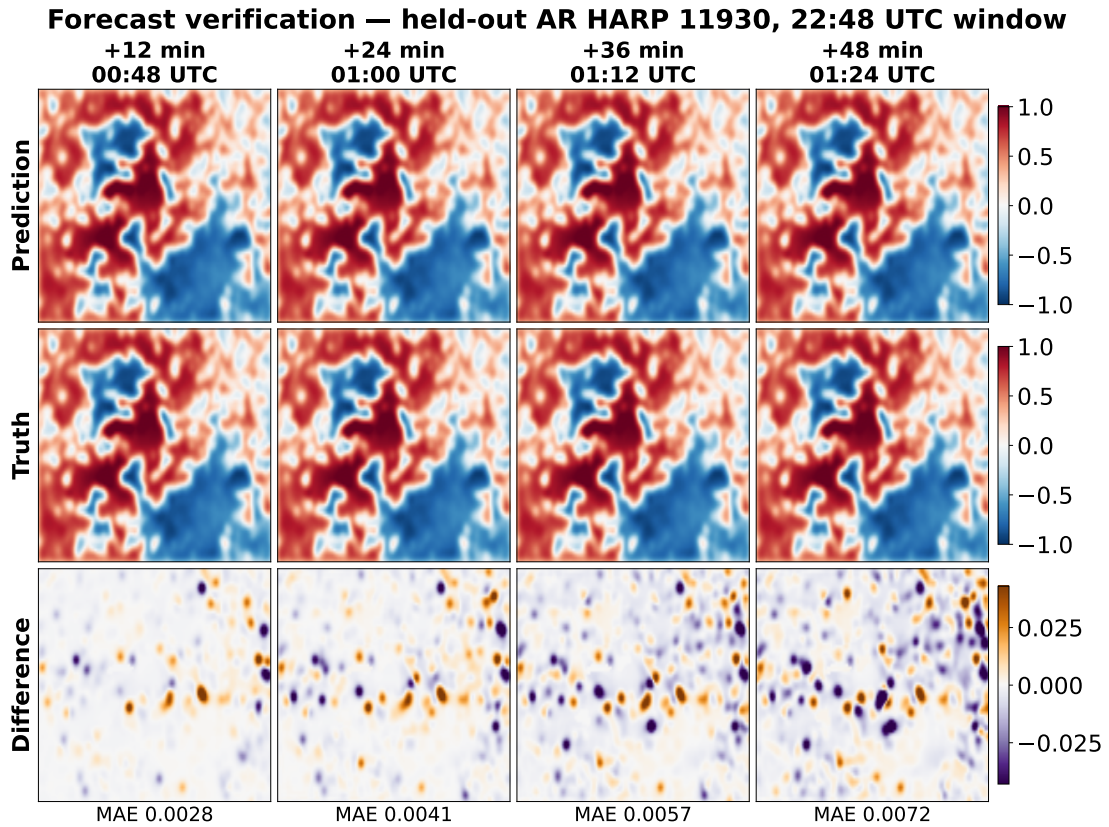


Figure 3.1: Frame-by-frame verification on the held-out region. For one window of HARP 11930: predicted field (top), observed field (middle), and their difference (bottom, on an expanded colour scale) at lead times +12 to +48 minutes. Errors are small, grow with lead time (per-frame MAE annotated), and localise along the sharp-gradient boundaries between winding lobes—the fine structure that cannot be forecast.

3.3 Discussion

Three points must be made plainly, because the visual quality of Figure 3.3 is easy to over-read.

The skill is real but small, and persistence-dominated. The model beats persistence by 9.3% on a held-out region, so it has learned something transferable about how the winding field evolves, beyond copying. But the margin is modest, and it must be. As established in §2.1 and Figure 2.5, the accumulated total winding builds up gradually and is therefore strongly autocorrelated from frame to frame; persistence is already a strong forecast, and there is limited headroom above it. The striking agreement between prediction and observation in Figure 3.3 is therefore in large part a statement about the *predictability of the total field*, and the honest measure of what the model adds is the skill relative to persistence, not the visual match. Quoting that baseline alongside every result is what keeps the account truthful.

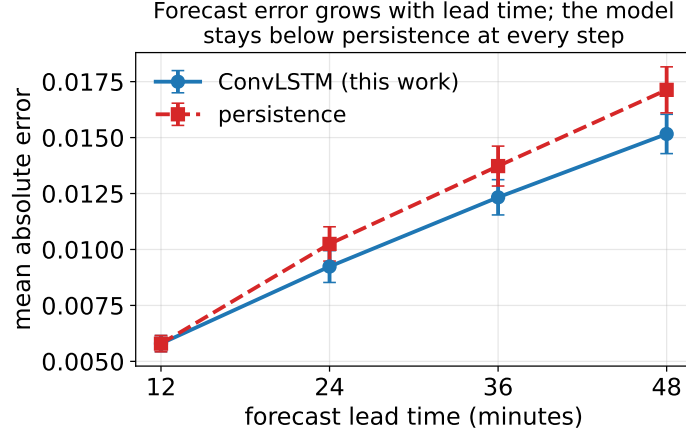


Figure 3.2: Forecast error versus lead time. Mean absolute error over all 61 held-out windows for the ConvLSTM and for persistence (error bars: standard error). The two coincide at +12 minutes and diverge thereafter, the model staying below persistence at every subsequent step.

Winding-flux forecast — held-out AR HARP 11930 ($t_{in} = 10 \rightarrow t_{out} = 4$, 12-min cadence)

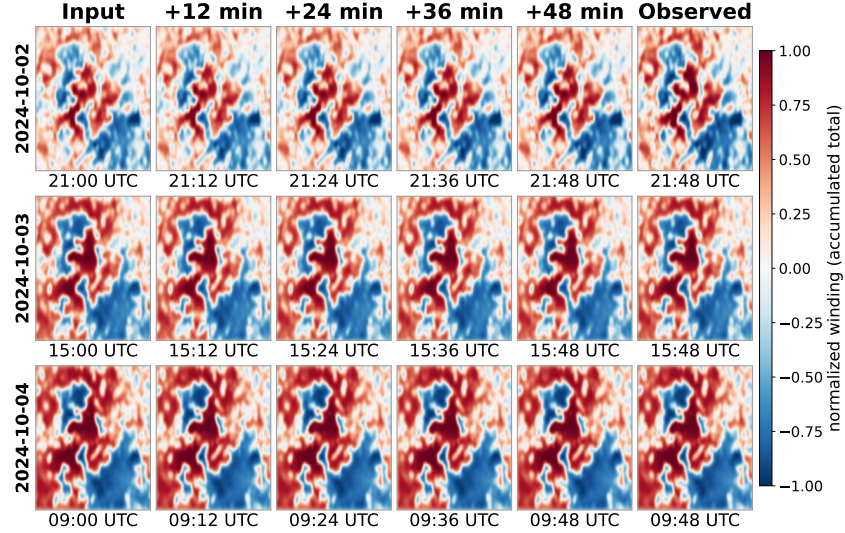


Figure 3.3: Winding-flux forecasts on the held-out active region HARP 11930. Each row is an independent window: the latest input frame, the four autoregressive forecasts at +12, +24, +36, +48 minutes, and the observed frame at +48 minutes, all stamped with their true UTC time. The model is trained only on the other 26 regions.

The forecastable content is the low-pass envelope. The spatial low-pass of §2.1 was applied deliberately, and the result vindicates the reasoning behind it: what the model reproduces is the large-scale envelope of the winding field, not its fine structure. The small-scale detail that the low-pass removed evolves incoherently from frame to frame and could not have been forecast in any case; excluding it lets the model spend its capacity where skill is attainable. This is an honest restriction of scope, not a hidden one.

Outlook. Taken together, the experiment delivers a working spatiotemporal pipeline for two-dimensional winding-flux maps and a clear-eyed measurement of what such

forecasting can yield. The path from here is not a better frame predictor—the persistence ceiling makes that a diminishing return—but the use of the learned spatiotemporal representation for the task that motivated the winding map in the first place: discriminating flare-productive from quiet configurations from the structure the scalar integral throws away.

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